Multilevel Modeling

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1. Introduction to Multilevel Modeling

1.1. Overview

Multilevel modeling, also known as hierarchical linear modeling (HLM) or mixed-effects modeling, is a statistical technique used to analyze data that have a hierarchical or nested structure. This approach is particularly useful when dealing with data that exhibit dependencies or correlations among observations at multiple levels or when there is a need to account for variability at different levels of a data hierarchy.

In multilevel modeling, data are typically organized into multiple levels, such as individuals within groups, students within schools, or patients within hospitals. The key idea is to recognize that observations within the same group or cluster are likely to be more similar to each other than to observations in other groups. Multilevel models allow researchers to account for this within-group correlation and estimate the effects of predictors at different levels of the hierarchy.

1.2. Key Concepts and Features

- Levels: Multilevel models have at least two levels, often referred to as the "lower" and "higher" levels. For example, in educational research, students (level 1) may be nested within classrooms (level 2), which are in turn nested within schools (level 3).
- Random Effects: Multilevel models include random effects at the higher levels to capture the variability between groups. These random effects are often represented as random intercepts and random slopes, allowing for group-specific variations in the outcome variable and the effects of predictors.
- Fixed Effects: Fixed effects represent the relationships between predictors and the outcome variable at each level. These are similar to the coefficients in traditional linear regression models.

- Hierarchical Structure: Multilevel models explicitly account for the hierarchical structure of the data by considering both within-group and between-group variations. This helps in modeling and understanding how individual and group-level factors influence the outcome.
- Modeling Variability: Multilevel modeling allows for the estimation of variability at each level of the hierarchy. Researchers can assess the proportion of variance in the outcome explained by each level, helping to identify where most of the variation occurs.
- Nested Data: Multilevel modeling is particularly useful when dealing with nested or clustered data, where observations within the same group are more correlated than those in different groups.
- Cross-Level Interactions: Multilevel models can also handle interactions between predictors at different levels, allowing researchers to explore how group-level factors may moderate the effects of individual-level predictors.

Multilevel modeling is widely used in various fields, including education, psychology, epidemiology, and social sciences, where data often exhibit nested structures. It provides a flexible and powerful framework for analyzing complex data with hierarchical dependencies and can yield more accurate and nuanced insights compared to traditional linear models that do not account for these dependencies.

2. {lme4} Basics

2.1. Overview of {lme4}

The {lme4} package provides functions for conducting various multilevel modeling techniques, including linear ({lmer}), generalized linear ({glmer}), and nonlinear ({nlmer}) models.

The 'official' tutorial is available here. More technically inclined readers may also find this article useful.

2.2. Setting up Environment

```
### Set the 'Cloud' mirror as a default CRAN (Comprehensive R Archive Network) repository
### CRAN is a global network of web servers that store identical, up-to-date versions of code and docum
options(repos="https://cloud.r-project.org")
```

3. Intraclass Correlation Coefficient

Intraclass correlation coefficient (ICC) is a statistical measure used in the context of multilevel or hierarchical data to quantify the proportion of total variance in a dependent variable that is attributable to the differences between groups or clusters.

In practice, ICC values between 0 and 1 suggest varying degrees of dependence or similarity among observations within the same group. A higher ICC indicates that a larger proportion of the total variance is due to group differences, implying a stronger clustering effect.

Researchers use the ICC to assess the appropriateness of using multilevel models to analyze their data. If the ICC is substantial, it suggests that accounting for group-level variability is important in modeling the data, as ignoring it may lead to biased or inefficient parameter estimates.

```
### Install and load the package
install.packages(c("lme4", "lmerTest"))
library(lme4)
library(lmerTest)
```

```r
### <Dyestuff> data
head(Dyestuff)
```

```
##   Batch Yield
## 1     A  1545
## 2     A  1440
## 3     A  1440
## 4     A  1520
## 5     A  1580
## 6     B  1540
```

```r
### Null model
null <- lmer(Yield ~ 1 + (1|Batch), data = Dyestuff, REML = FALSE)
summary(null)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
##   method [lmerModLmerTest]
## Formula: Yield ~ 1 + (1 | Batch)
##    Data: Dyestuff
##
##      AIC      BIC   logLik deviance df.resid
##    333.3    337.5   -163.7    327.3       27
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.4315 -0.7972  0.1480  0.7721  1.8037
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  Batch    (Intercept) 1388     37.26
##  Residual             2451     49.51
## Number of obs: 30, groups:  Batch, 6
##
## Fixed effects:
##             Estimate Std. Error     df t value Pr(>|t|)
## (Intercept)   1527.5       17.7    6.0   86.33 1.63e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

4. Adding Level-1 Predictors

```r
### Read the .dat data
dat <- read.table(file.choose(), header = TRUE) # Choose <popular2.dat> file

### Null model
null <- lmer(popular ~ 1 + (1|class), data = dat)
summary(null)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
```

```
## Formula: popular ~ 1 + (1 | class)
##    Data: dat
##
## REML criterion at convergence: 6330.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.5655 -0.6975  0.0020  0.6758  3.3175
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  class    (Intercept) 0.7021   0.8379
##  Residual             1.2218   1.1053
## Number of obs: 2000, groups:  class, 100
##
## Fixed effects:
##             Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)  5.07786    0.08739 98.90973    58.1   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
confint(null, method = "profile", level = 0.95)
```

```
## Computing profile confidence intervals ...
```

```
##                2.5 %    97.5 %
## .sig01      0.719963 0.9744636
## .sigma      1.071113 1.1414463
## (Intercept) 4.905780 5.2499439
```

```r
### Add level-1 predictor 'extrav': random intercept, fixed slope
l1_fs <- lmer(popular ~ 1 + extrav + (1|class), data = dat)
summary(l1_fs)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: popular ~ 1 + extrav + (1 | class)
##    Data: dat
##
## REML criterion at convergence: 5832.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.0644 -0.7267  0.0165  0.7088  3.3587
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  class    (Intercept) 0.8406   0.9168
##  Residual             0.9304   0.9646
## Number of obs: 2000, groups:  class, 100
##
## Fixed effects:
```

```
##              Estimate Std. Error        df t value Pr(>|t|)
## (Intercept) 2.542e+00  1.411e-01 4.380e+02   18.01   <2e-16 ***
## extrav      4.863e-01  2.015e-02 1.965e+03   24.13   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##        (Intr)
## extrav -0.745
```

```r
confint(l1_fs, method = "profile", level = 0.95)
```

```
## Computing profile confidence intervals ...
```

```
##                 2.5 %    97.5 %
## .sig01      0.7913536 1.0624251
## .sigma      0.9344460 0.9958162
## (Intercept) 2.2647455 2.8187560
## extrav      0.4465543 0.5258169
```

```r
## Add level-1 predictor 'extrav': random intercept, random slope
l1_rs <- lmer(popular ~ 1 + extrav + (1 + extrav|class), data = dat)
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge with max|grad| = 0.00663866 (tol = 0.002, component 1)
```

```r
summary(l1_rs)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: popular ~ 1 + extrav + (1 + extrav | class)
##    Data: dat
##
## REML criterion at convergence: 5779.4
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.1961 -0.7291  0.0145  0.6817  3.2216
##
## Random effects:
##  Groups   Name        Variance Std.Dev. Corr
##  class    (Intercept) 2.99680  1.7311
##           extrav      0.02595  0.1611   -0.97
##  Residual             0.89495  0.9460
## Number of obs: 2000, groups:  class, 100
##
## Fixed effects:
##             Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)  2.46106    0.20309 96.71010   12.12   <2e-16 ***
## extrav       0.49286    0.02545 89.75489   19.36   <2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##        (Intr)
## extrav -0.917
## optimizer (nloptwrap) convergence code: 0 (OK)
## Model failed to converge with max|grad| = 0.00663866 (tol = 0.002, component 1)
```

```r
confint(l1_rs, method = "profile", level = 0.95)
```

```
## Computing profile confidence intervals ...
```

```
##                  2.5 %     97.5 %
## .sig01       1.4156977  2.0829902
## .sig02      -1.0000000 -0.8989131
## .sig03       0.1056739  0.2170604
## .sigma       0.9159564  0.9778293
## (Intercept)  2.0583720  2.8594566
## extrav       0.4422616  0.5429940
```

5. Comparing Models

Likelihood-ratio test (LRT) is a statistical method used for comparing two nested statistical models, typically in the context of hypothesis testing. It is a common tool in statistics and econometrics and is used to determine whether a more complex model significantly improves the fit of the data compared to a simpler model. LRT is based on the likelihood function, which measures how well a statistical model explains the observed data.

- The likelihood ratio statistic (LR) is calculated as the ratio of the likelihood of the data under the more complex model (H1) to the likelihood of the data under the simpler model (H0).
- If LR is greater than the critical value, you reject the null hypothesis (H0) in favor of the alternative hypothesis (H1), indicating that the more complex model provides a significantly better fit to the data.
- If LR is not greater than the critical value, you fail to reject the null hypothesis, suggesting that the simpler model is sufficient to explain the data.

```r
## LRT
anova(null, l1_fs)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: dat
## Models:
## null: popular ~ 1 + (1 | class)
## l1_fs: popular ~ 1 + extrav + (1 | class)
##       npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
## null     3 6333.5 6350.3 -3163.7   6327.5
## l1_fs    4 5831.8 5854.2 -2911.9   5823.8 503.69  1  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(l1_fs, l1_rs)
```

```
## refitting model(s) with ML (instead of REML)

## Data: dat
## Models:
## l1_fs: popular ~ 1 + extrav + (1 | class)
## l1_rs: popular ~ 1 + extrav + (1 + extrav | class)
##       npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
## l1_fs    4 5831.8 5854.2 -2911.9   5823.8
## l1_rs    6 5782.7 5816.3 -2885.3   5770.7 53.089  2  2.964e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

6. Adding Level-2 Predictors

```r
### Add level-2 predictor 'texp'
l1_rs_l2 <- lmer(popular ~ 1 + extrav + texp + (1 + extrav|class), data = dat)
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge with max|grad| = 0.00290535 (tol = 0.002, component 1)
```

```r
summary(l1_rs_l2)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: popular ~ 1 + extrav + texp + (1 + extrav | class)
##    Data: dat
##
## REML criterion at convergence: 5749.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.0473 -0.7305  0.0110  0.6777  3.2031
##
## Random effects:
##  Groups   Name        Variance Std.Dev. Corr
##  class    (Intercept) 1.56077  1.2493
##           extrav      0.03535  0.1880   -0.87
##  Residual             0.89057  0.9437
## Number of obs: 2000, groups:  class, 100
##
## Fixed effects:
##              Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)   1.01714    0.23075 191.13070   4.408 1.74e-05 ***
## extrav        0.50215    0.02764  96.48459  18.171  < 2e-16 ***
## texp          0.09774    0.01026 100.55894   9.527 1.03e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Correlation of Fixed Effects:
##        (Intr) extrav
## extrav -0.719
## texp   -0.702  0.103
## optimizer (nloptwrap) convergence code: 0 (OK)
## Model failed to converge with max|grad| = 0.00290535 (tol = 0.002, component 1)
```

```r
confint(l1_rs_l2, method = "profile", level = 0.95)
```

```
## Computing profile confidence intervals ...
```

```
##                   2.5 %     97.5 %
## .sig01       0.93333274  1.5794576
## .sig02      -0.93028020 -0.7401041
## .sig03       0.13175425  0.2446509
## .sigma       0.91377693  0.9752509
## (Intercept)  0.50462689  1.5506855
## extrav       0.44769468  0.5566324
## texp         0.06980318  0.1251256
```

```r
### LRT
anova(l1_rs, l1_rs_l2)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: dat
## Models:
## l1_rs: popular ~ 1 + extrav + (1 + extrav | class)
## l1_rs_l2: popular ~ 1 + extrav + texp + (1 + extrav | class)
##          npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
## l1_rs       6 5782.7 5816.3 -2885.3   5770.7
## l1_rs_l2    7 5747.0 5786.2 -2866.5   5733.0 37.658  1  8.429e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
## Add cross-level interaction
l1_rs_l2_int <- lmer(popular ~ 1 + extrav + texp + extrav*texp + (1 + extrav|class), data = dat)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
summary(l1_rs_l2_int)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: popular ~ 1 + extrav + texp + extrav * texp + (1 + extrav | class)
##    Data: dat
##
## REML criterion at convergence: 5694.8
##
## Scaled residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -3.2505 -0.7267  0.0162  0.6861  3.1895
##
## Random effects:
##  Groups   Name        Variance  Std.Dev. Corr
##  class    (Intercept) 0.5290163 0.72734
##           extrav      0.0003338 0.01827  -1.00
##  Residual             0.8898021 0.94329
## Number of obs: 2000, groups:  class, 100
##
## Fixed effects:
##               Estimate Std. Error         df t value Pr(>|t|)
## (Intercept) -1.211e+00  3.210e-01  1.860e+02  -3.772 0.000218 ***
## extrav       8.911e-01  4.495e-02  1.759e+03  19.822  < 2e-16 ***
## texp         2.519e-01  1.975e-02  1.660e+02  12.758  < 2e-16 ***
## extrav:texp -2.746e-02  2.872e-03  1.725e+03  -9.563  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) extrav texp
## extrav      -0.874
## texp        -0.918  0.805
## extrav:texp  0.773 -0.898 -0.859
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```r
confint(l1_rs_l2_int, method = "profile", level = 0.95)
```

```
## Computing profile confidence intervals ...

## Warning in FUN(X[[i]], ...): non-monotonic profile for .sig02

## Warning in nextpar(mat, cc, i, delta, lowcut, upcut): unexpected decrease in
## profile: using minstep

## Warning in FUN(X[[i]], ...): non-monotonic profile for .sig03

## Warning in confint.thpr(pp, level = level, zeta = zeta): bad spline fit for
## .sig02: falling back to linear interpolation

## Warning in confint.thpr(pp, level = level, zeta = zeta): bad spline fit for
## .sig03: falling back to linear interpolation

##                   2.5 %      97.5 %
## .sig01       0.49745999  0.97054491
## .sig02      -1.00000000  1.00000000
## .sig03       0.00000000  0.10027043
## .sigma       0.91361314  0.97361459
## (Intercept) -1.83975392 -0.57950433
## extrav       0.80302901  0.97918136
## texp         0.21299762  0.29058171
## extrav:texp -0.03309267 -0.02182997
```

```r
### LRT
anova(l1_rs_l2, l1_rs_l2_int)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: dat
## Models:
## l1_rs_l2: popular ~ 1 + extrav + texp + (1 + extrav | class)
## l1_rs_l2_int: popular ~ 1 + extrav + texp + extrav * texp + (1 + extrav | class)
##              npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
## l1_rs_l2        7 5747.0 5786.2 -2866.5   5733.0
## l1_rs_l2_int    8 5683.9 5728.7 -2834.0   5667.9 65.117  1  7.058e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
## Remove level-1 slope variance
l1_fs_l2_int <- lmer(popular ~ 1 + extrav + texp + extrav*texp + (1|class), data = dat)
summary(l1_fs_l2_int)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: popular ~ 1 + extrav + texp + extrav * texp + (1 | class)
##    Data: dat
##
## REML criterion at convergence: 5695.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.2330 -0.7279  0.0079  0.6964  3.1814
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  class    (Intercept) 0.3960   0.6293
##  Residual             0.8906   0.9437
## Number of obs: 2000, groups:  class, 100
##
## Fixed effects:
##               Estimate Std. Error         df t value Pr(>|t|)
## (Intercept) -1.219e+00  3.090e-01  7.614e+02  -3.945 8.72e-05 ***
## extrav       8.924e-01  4.495e-02  1.954e+03  19.851  < 2e-16 ***
## texp         2.527e-01  1.895e-02  7.253e+02  13.333  < 2e-16 ***
## extrav:texp -2.759e-02  2.869e-03  1.984e+03  -9.618  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) extrav texp
## extrav      -0.857
## texp        -0.919  0.791
## extrav:texp  0.758 -0.899 -0.842
```

```r
confint(l1_fs_l2_int, method = "profile", level = 0.95)
```

```
## Computing profile confidence intervals ...
```

```
##                   2.5 %     97.5 %
## .sig01       0.53506603  0.7297867
## .sigma       0.91402218  0.9740407
## (Intercept) -1.82301015 -0.6149334
## extrav       0.80436473  0.9804347
## texp         0.21561240  0.2897011
## extrav:texp -0.03321434 -0.0219763
```

```r
### LRT
anova(l1_fs_l2_int, l1_rs_l2_int)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: dat
## Models:
## l1_fs_l2_int: popular ~ 1 + extrav + texp + extrav * texp + (1 | class)
## l1_rs_l2_int: popular ~ 1 + extrav + texp + extrav * texp + (1 + extrav | class)
##              npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
## l1_fs_l2_int    6 5680.8 5714.4 -2834.4   5668.8
## l1_rs_l2_int    8 5683.9 5728.7 -2834.0   5667.9 0.8558  2     0.6519
```